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Database & MySQL TEST (1)

1. What is Data?

- ⇒ (i) Data is raw information or values.
- (ii) It can be stored and processed.
- (iii) Also, it is an data like in numbers or characters etc.

2. What is a database?

- ⇒ (i) An organized collection of related data stored systematically so it can be easily accessed, managed and updated.

(ii) Ex University

college
students
lectures
clerk

3. What does DBMS stands for?

⇒ (i) Data Base Management system

- (ii) It uses for managing the biggest data in a worst situation.

4) What does RDBMS stand for?

⇒ (i) Relational Database Management system.

5. What is MySQL?

- ⇒ (i) MySQL is an open-source.
- (ii) It's known as My structured Query Language.
- (iii) It's functions as a application that stores, organizes & data using tables in format of columns & rows.

6. What is SQL?

- ⇒ (i) It known as structured query language.
- (ii) It's standard language used to communicate with & manage databases.

7. Name any two database models.

- ⇒ (i) Hierarchical model
- (ii) Relational Model

8. What is a row in a database table?

- ⇒ (i) A row is a single record in a table.
- (ii) It also a tuple. & contains data for columns of that table.

9. What is the purpose of Distinct keyword?

- ⇒ (i) Distinct is used to remove duplicate values.
- (ii) Show only unique values.

(3)

10. What does the % wildcard represent in MySQL?

- ⇒ (i) % represent zero.
(ii) Multiple characters like 'A%' and all names starting with A.

B-02
11. Differentiate between Database & DBMS.

Database	DBMS
It is the collection of data.	It is the software to manage the database.
Does not provide security itself.	Provide security, backup, & recovery.
It depends on the data which we use in it.	It depends on the database which is used in it.
Eg. file with student data.	eg. MySQL, Oracle.

12. Differentiate between SQL & MySQL.

SQL	MySQL
It is language.	It is a software and DBMS.
(i) used to query database.	(ii) uses SQL to perform operations.
(ii) It is a standard.	(iii) It is a product that implement SQL.
(iv) It's communicate with relational database.	(iv) It uses on DBMS and as well in SQL.

13. Write any four SQL commands categories. & gives one command each.

- DDL (Data Definition Language):
Create Table students (...);
- DML (Data Manipulation Language):
Insert into student values (...);
- DQL (Data Query Language):
select * from students;
- DCL (Data Control Language):
Grant select on students to user1;

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14) What is the difference between IN & BETWEEN?

IN operator

- (i) It checks if a value matches any value in a given list.
- (ii) Select from students where city IN ('Pune', 'Mumbai');

BETWEEN operator

- (i) Checks if a value is within a range.
- (ii) Select from students WHERE marks BETWEEN 60 and 90;

15) Explain NULL and IS NOT NULL with suitable SQL examples.

I. IS NULL :-

Finds records where a column has no value or empty.

select from students where Email IS NULL;

II. IS NOT NULL :-

Finds records where a column has a value.

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Select * from students where email is not null;

16) Explain DBMS and RDBMS. Write & differentiate them.

DBMS: A software to store and manage data in files. Data may be redundant. No relationship between tables is strictly enforced. eg. file systems, ms Access (older).

RDBMS: A type of DBMS that stores data in table (relations) and manages relationships between them using keys. More secure & powerful. eg. mysql, Oracle.

DBMS	RDBMS
(i) Stores data as files	Stores data in tables with rows & columns.
(ii) No or less data integrity constraints.	Supports integrity constraints (primary key, foreign key)
(iii) Data redundancy is high.	Redundancy is less due to Normalization.
(iv) Does not support relationships well	Supports table relationships.

Structure: Flexible.

9) Explain the following database models with their basic structure.

- Hierarchical Database**: Data organized like a tree, parent-child relationship. One parent can have many children, but a child has only one parent.
Eg: File system. Structure: Tree.
- Relational Database**: Most popular model. Data stored in tables (relations). Tables are related using Primary key & Foreign key. Structure: Tables.
- Network Database**: More flexible than hierarchical. A child can have multiple parents. Use many-to-many relationship.
Structure: Graph, Network.
- Object-Oriented Database**: Data stored as object like in OOP programming. An object contains data, & methods. Complex data like images, video.
Structure: Objects.
- NoSQL Database**: Non-relational. Does not use fixed tables. Used for big data, unstructured data. Types: Document, Key-Value, Graph, Column, e.g. MongoDB.
Structure: Flexible.

10) Explain the six MySQL constraints given in the notes. Write the purpose of each.

- NOT NULL**: Ensure column cannot have a NULL (empty) value.
name varchar(50) NOT NULL
- UNIQUE**: Ensure all values in a column are different - no duplicates.
email varchar(100) UNIQUE
- Default**: sets a default value for a column if no value is provided.
city varchar(50) default('Pune')
- Check**: Ensure values in a column satisfy a specific condition.
age INT (CHECK (age >= 18))
- Primary key** & **Foreign key**:
Used to link two tables; ensure referential integrity. Values must exist in the parent table.
Foreign key (dept_id) references department (id)

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6. Primary key :-
Combines NOT Null & Unique.
Uniquely identifies each row. One
per table.
id int primary key.

19. student.

a. display ... age is 20.

=> select * from student where age = 20;

b. Display .. age between 20 & 23.

=> select * from student where age between 20 and 23;

c. Display students .. Delhi & Mumbai.

=> select * from student where city IN ('Delhi', 'Mumbai');

d. Display only female students.

=> select * from student where gender = 'female';

e. Display students whose ... start with P.

=> select * from student where name like 'P%';

20. my sql like operator & its wild card o/p and ..

=> like operator is used for pattern matching in where clause.

two wild cards

1. % (Percent) : represent zero, one, or multiple characters.

2. _ (Underscore) : represent a single character.

a. Name starting with A.

=> select * from student where name like 'A%';

b. ending ... a

=> select * from student where name like '%a';

c. Name ... an

=> select * from student where name like '%an%';

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(d) ... second character.

⇒ select * from student where like ' %'

(e) starting o as third character.

⇒ select * from student where
name like - o %;